

2.3 Algorithms

The use of algorithms to describe problems and standard algorithms

2.3.1 Algorithms

- (a) Analysis and design of algorithms for a given situation.
- (b) The suitability of different algorithms for a given task and data set, in terms of execution time and space.
- (c) Measures and methods to determine the efficiency of different algorithms, Big O notation (constant, linear, polynomial, exponential and logarithmic complexity).
- (d) Comparison of the complexity of algorithms.
- (e) Algorithms for the main data structures, (stacks, queues, trees, linked lists, depth-first (post-order) and breadth-first traversal of trees).
- (f) Standard algorithms (bubble sort, insertion sort, merge sort, quick sort, Dijkstra's shortest path algorithm, A* algorithm, binary search and linear search).